

BCA 1st Sem. (2019-22) Maths Foundation

Time : 3 hours

Full Marks : 80

Note: Candidates are required to give their answers in their own words as far as practicable. The questions are of equal value. Answer any five questions.

1. Find the Eigen values and Eigen Vectors of the following matrix:

(a) $A = \begin{bmatrix} 1 & 1 \\ 3 & -1 \end{bmatrix}$

(b) $B = \begin{bmatrix} 3 & 2 & 2 \\ 2 & 2 & 0 \\ 2 & 0 & 4 \end{bmatrix}$

2. Find the inverse of the matrix: $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 2 & 4 \\ 6 & 3 & -1 \end{bmatrix}$

3. Find the rank of the matrix: $A = \begin{bmatrix} 1 & 2 & -3 \\ -1 & 2 & 1 \\ -2 & 1 & -2 \end{bmatrix}$

4. Find the $\max_m$ and $\min_m$ of the function:

$$f(x, y) = x^3 + y^3 - 3x - \frac{1}{y^2}$$

5. (a) If $Z = \tan^{-1} \left(\frac{2xy}{x^2 - y^2} \right)$.

Prove that $\frac{\partial Z}{\partial x} + \frac{\partial Z}{\partial y} = 0$.

- (b) If $u = \sin^{-1} \left(\frac{x}{y} \right)$.

Prove that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = 0$.

6. Solve: $(x^2 + y^2) \frac{dy}{dx} = 2xy$

7. Integrate: $(x + y)^2 \frac{dy}{dx} = 2(x + y)^2 + 3 = 0$

8. Evaluate the integrals:

(a) $I = \int_0^{\frac{\pi}{2}} \sqrt{1 + \sin x} \, dx$

(b) $I = \int_0^2 \int_1^2 (x^2 + y^2) \, dx \, dy$

(c) $I = \int_0^1 \int_{x^2}^{2-x} xy \, dx \, dy$

9. Solve the following differential equation:

$$(x + y)^2 \frac{dy}{dx} = 2(x + y)^2 + 3 = 0$$

10. Write Short notes on:

(a) Inverse Matrix

(b) Transpose Matrix

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